

# Maximization of the operational multiplication factor in a photon driven subcritical reactor: a methodological approach for temperature and poison feedback

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# The Global Energy Challenge

## Climate Emergency

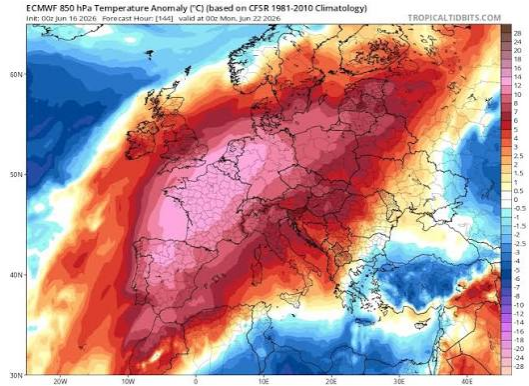
Record CO<sub>2</sub> emissions and global warming drive the urgent need for decarbonization. Most of worldwide electricity still relies on fossil fuels, while energy demand is constantly increasing.

## A Possible Solution

Transitioning to a **carbon-free baseload** requires innovative technologies.

**Accelerator-Driven Systems** provide a unique solution:

- Inherent safety due to subcritical operation
- Power modulation via beam intensity



Intense heat wave over Europe in the final days of June 2026.

# Thesis Context and Objective

## Subcritical System

A reactor with  $k_{eff} < 1$  requires an **external neutron source** to sustain the chain reaction. During operational transients, **feedback** modify  $k_{eff}$  .

## Objective

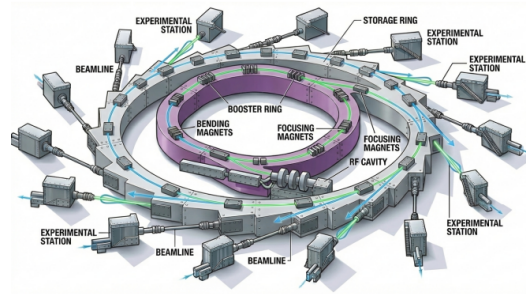
Develop a methodology to **maximize**  $k_{eff}^{HFP}$  (deeply linked to power production) ensuring that  $k_{eff}^{CZP} \leq 1$  under any operating condition.

- **CZP** (Cold Zero Power):  $T = 300$  K — reactor shutdown
- **HFP** (Hot Full Power):  $T_{fuel} = 600$  K,  $T_{mod} = 350$  K — nominal operation

# SYNERGY Project: Photonic Source

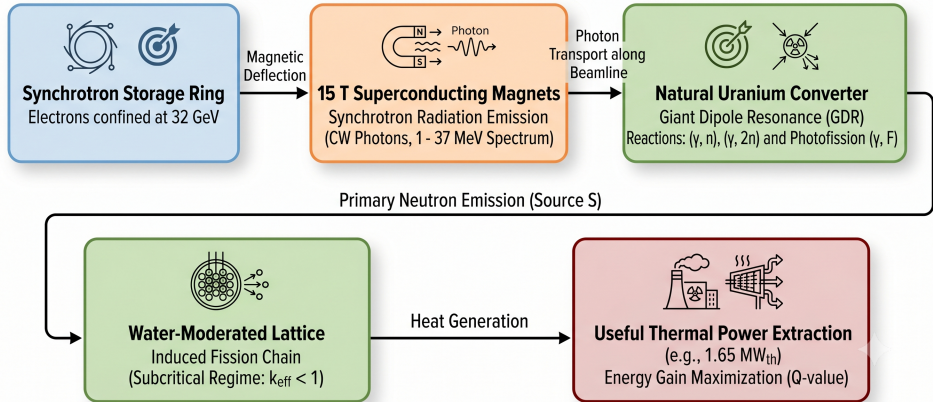
To overcome the limitations of accelerator based neutron sources — specifically the excessive heat deposition of charged particles on the converter — this thesis is developed within the SYNERGY project:

- Synchrotron-based neutron source utilizing extracted  $\gamma$ -photons (1–37 MeV) to mitigate thermal load on the converter
- Photonuclear ( $\gamma, xn$ ) reactions and photofission ( $\gamma, f$ ) on a natural Uranium target
- Photonic rate:  $S_\gamma = 2.02 \times 10^{17} \text{ } \gamma/\text{s}$
- Neutronic rate up to  $10^{15} \text{ n/s}$
- Beam power:  $\sim 200 \text{ kW}$  per beamline



# From Source to Core: The Physical Workflow

## SYNERGY PROJECT: ACCELERATOR-DRIVEN SYSTEM (ADS) PHYSICAL WORKFLOW & PARTICLE CASCADE



# Reactivity Feedback

During the transition from HFP to CZP, the temperature drop induces a **reactivity insertion**. The primary feedback mechanisms are:

## Doppler Effect

- Narrowing of  $^{238}\text{U}$  capture resonances due to reduced thermal agitation
- Decreased epithermal absorption  $\Rightarrow \Delta\rho > 0$

## Moderator Effect

- Increase in water density as temperature decreases
- Enhanced neutron thermalization  $\Rightarrow \Delta\rho > 0$

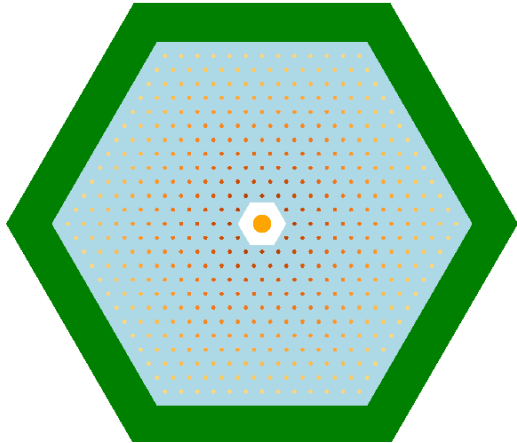
### Reactivity Variation

$$\Delta\rho = \frac{k_2 - k_1}{k_1 \cdot k_2} \quad [\text{pcm}]$$

### Safety Constraint

$$k_{max}^{HFP} \approx k_{max}^{CZP} - \sum_i \Delta\rho_i$$

# Core Topology and Variable Enrichment



*Hexagonal Lattice (XY Section)*

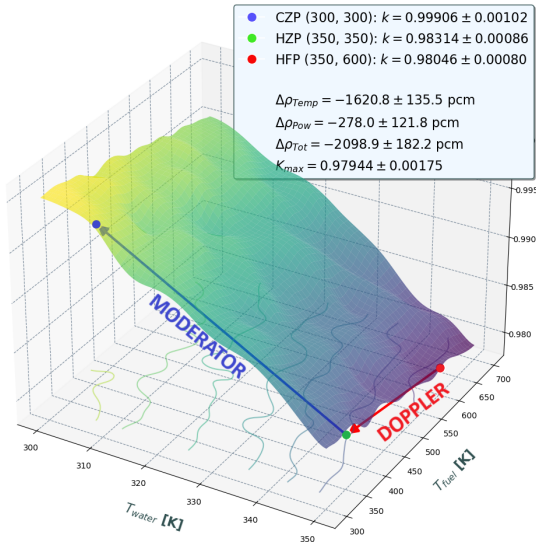
## Core Materials

- **Fuel:**  $UO_2$  pins with Zirconium cladding.
- **Moderator:** Light water doped with 15% heavy water ( $H_2O + 15\% D_2O$ ).
- **Reflector:** Nuclear graphite.

## Enrichment Strategy (Fuel Zoning)

**Radial linear profile:**  $^{235}U$  enrichment varies from the inner rings towards the core periphery.

# Thermodynamic Multiplication Surface



3D map of  $k_{eff}(T_{fuel}, T_{mod})$  in the HFP  $\rightarrow$  HZP  $\rightarrow$  CZP transient.

The system cooling causes a **reactivity increase** that must be compensated by the initial subcriticality margin.

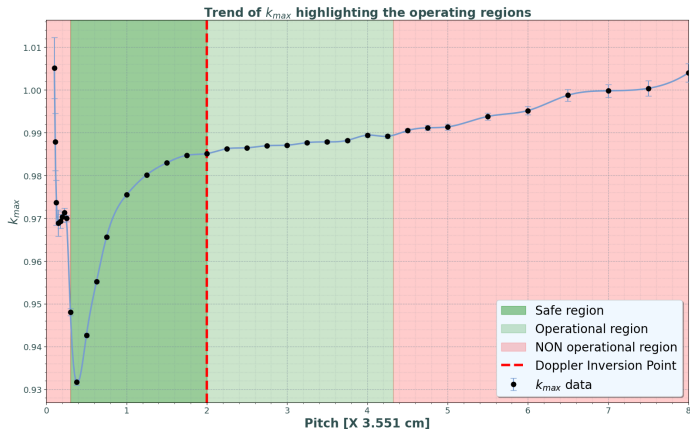
In addition, another 200 pcm are added as a safety margin to account for uncertainties in the feedback quantification.

## Pitch optimization

Keeping the operating temperatures fixed, what is the optimal pitch to maximize  $k_{max}$  ?

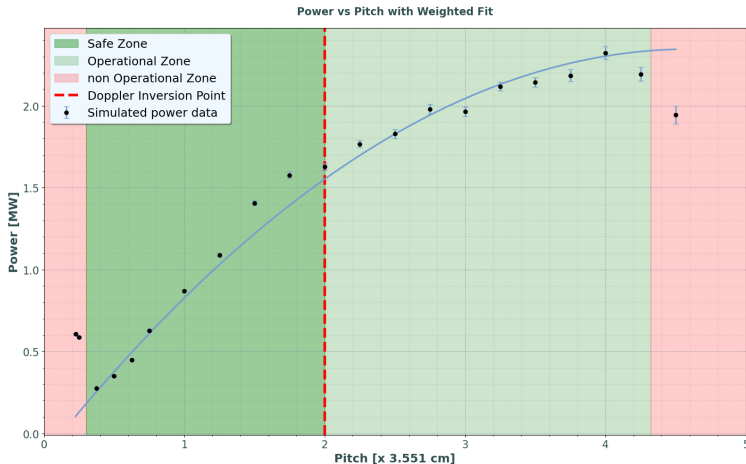
# Lattice Pitch Optimization

Pitch governates the moderation level and the neutron spectrum, thus affecting the multiplication factor



- **Green Zone:** Possible operation — target  $k_{max}$  reachable by modulating enrichment while the **Red Zone** is inaccessible,  $k_{max}$  is not reached even at 50% enrichment
- Important to emphasized that over multiplication pitch of 2 **the doppler effect inverts sign**, meaning that the reactor is instable at source oscillation

# Results: Power vs. Pitch



## Feasible Maximum

- Pitch  $\approx 7$  cm
- $P_{th} = 1.65$  MW
- **Stable and safe**
- $Q$ -Value = 4.12

$$Q = \frac{P_{Th}}{P_{El}/\eta_{Th-El}}$$

# Neutron Poisons: Xe-135 and Sm-149

Other than the thermal feedback, the presence of neutron poisons (fission products with high absorption cross section) can significantly reduce  $k_{eff}$  during operation. The main poisons in a uranium-fueled reactor are:

## Xenon-135

- $\sigma_a \approx 2 \times 10^6$  b
- Transient: **post-shutdown peak**
- Decays with  $T_{1/2} = 9.2$  h

$$N_{Xe}^{eq} = \frac{(\gamma_I + \gamma_{Xe})\Sigma_f\phi}{\lambda_{Xe} + \sigma_a\phi}$$

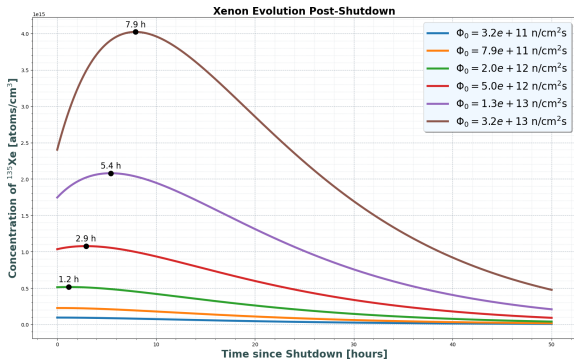
## Samarium-149

- $\sigma_a \approx 4.1 \times 10^4$  b
- **Stable** ( $\lambda_{Sm} = 0$ )
- Permanent asymptotic buildup

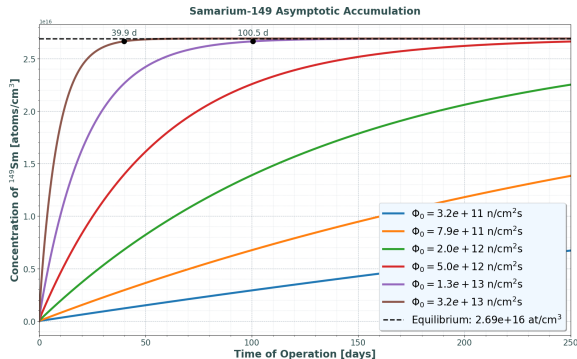
$$N_{Sm}^{eq} = \frac{\gamma_{Pm}\Sigma_f}{\sigma_{a,Sm}}$$

# Burnup Simulations: Poison Evolution

From the analytical resolution of the Bateman equations, it is possible to observe the behavior as the flux varies:

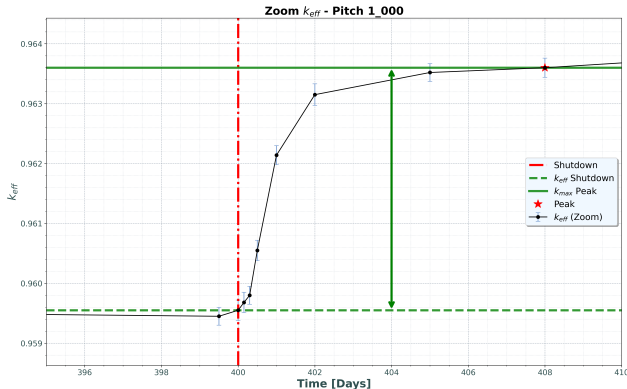


Xe-135: post-shutdown peak



Sm-149: asymptotic plateau

# Evolution of $k_{eff}$ over the Life Cycle

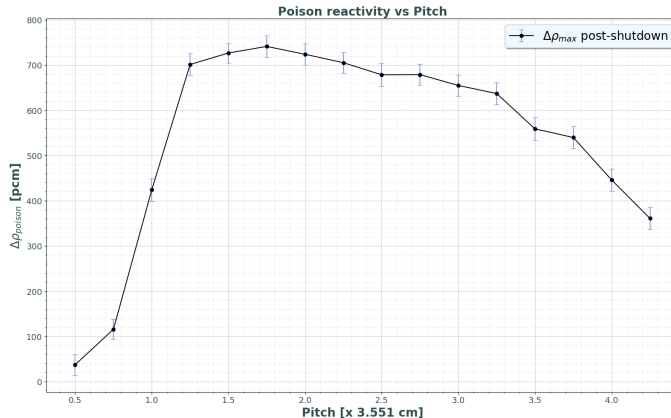


The post-shutdown  $k_{eff}$  evolution is driven by the net reactivity balance of neutron poisons:

- **Xenon-135** decays entirely without a transient peak, releasing positive reactivity.
- **Samarium-149** asymptotically accumulates, introducing a negative reactivity penalty.

Since the positive reactivity gained from Xenon's disappearance strictly outweighs the Samarium penalty, the system experiences a **net reactivity gain**, leading to the asymptotic growth of the fundamental eigenvalue.

# Poisoning Reactivity vs Pitch



**Figure:** Evolution of  $\Delta\rho_{\text{Poison}}$  as the Pitch varies. The analysis highlights the absence of a global optimum, identifying instead a limiting condition ("worst-case") near the x2 multiplier.

# Total Reactivity Balance

$$\Delta\rho_{tot} = \Delta\rho_{Doppler} + \Delta\rho_{Mod} + \Delta\rho_{Poison} + \Delta\rho_{Safety}$$

**Lattice Pitch: 3.55 cm**

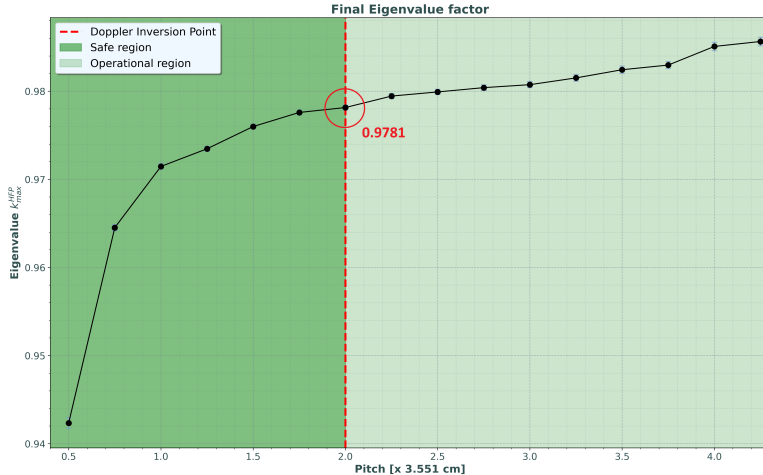
| <b>Contribution</b> | <b>[pcm]</b> |
|---------------------|--------------|
| Doppler             | 417          |
| Moderator           | 1897         |
| Poisons             | 424          |
| Safety              | 200          |
| <b>Total</b>        | <b>2938</b>  |

## Operational Limit

$$k_{max}^{HFP} \approx 1 - 2938 \text{ pcm} = \boxed{0.97062}$$

Below this limit, safety is guaranteed in any thermal transient.

# HFP Operability vs Pitch



**Figure:** Maximum  $k_{eff}$  value in HFP regime as the pitch varies. The trend shows a constant improvement towards pitch  $\times 4$ , where the highest available safe multiplication value is achieved.

## Methodological Framework

- **Parametric** and modular optimization procedure.
- **Intrinsically agnostic**: easily adaptable to any alternative reactor layout (geometry, fuel, reflector, or moderator).

## Quantitative Output vs. Lattice Pitch

- Maximum allowable operational eigenvalue ( $k_{\text{HFP}}^{\text{max}}$ ).
- Extracted thermal power and  $Q$ -value.
- Doppler, moderator density, and poisoning reactivity feedback.

## Original Contribution

First systematic quantification of the **total reactivity load** in a photon-driven ADS system through fully coupled Monte Carlo and depletion simulations (OpenMC + CRAM).

- **Thermal-Hydraulic Coupling:** 3D spatial integration between OpenMC and CFD/sub-channel solvers for exact local temperature gradient evaluation.
- **Dynamic Beam Control:** Implementation of a feedback control loop on the photon source intensity  $S(t)$  to compensate for poison accumulation and maintain a constant thermal power.
- **Source Kinematic Refinement:** Integration of the real anisotropic angular distributions (peaked at  $90^\circ$  for GDR) of the photoneutron emission.
- **Fast Spectrum Extension:** Application of the framework to a fast ADS core (Lead/LBE cooled) to optimize Minor Actinides (MA) transmutation.

# Thank you for your attention!

*Questions?*

# Variable Enrichment Geometry

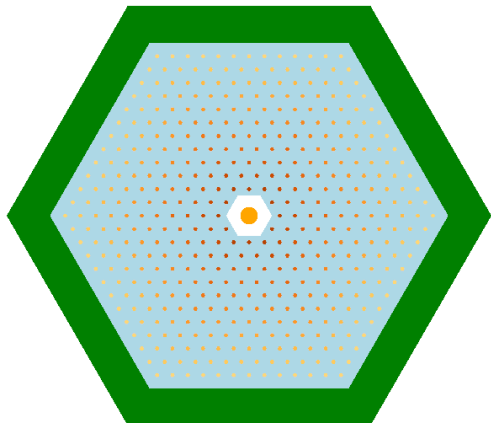


Figure: XY plane section.

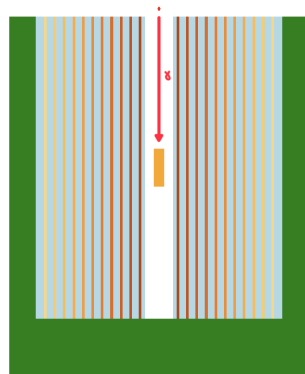
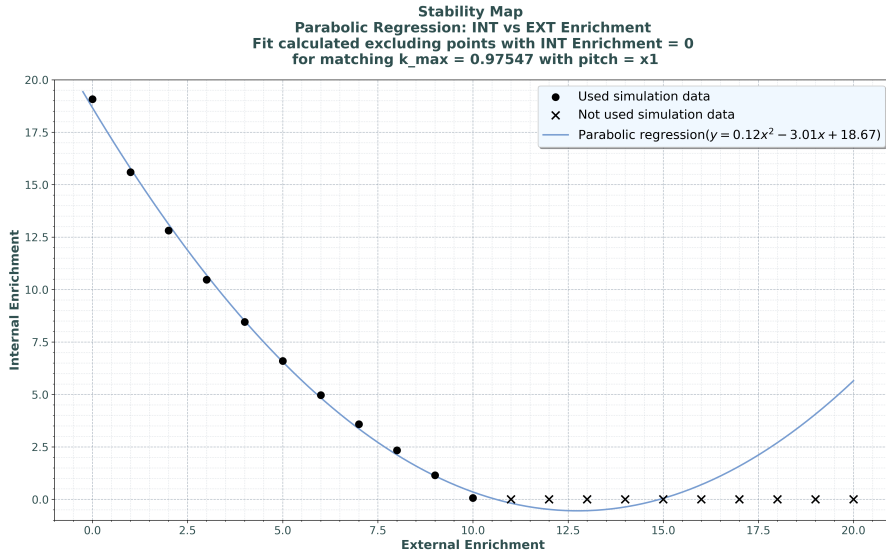


Figure: XZ plane section.

# Stable Solution Space for a 3.551 cm Pitch



# Stable Solution Space

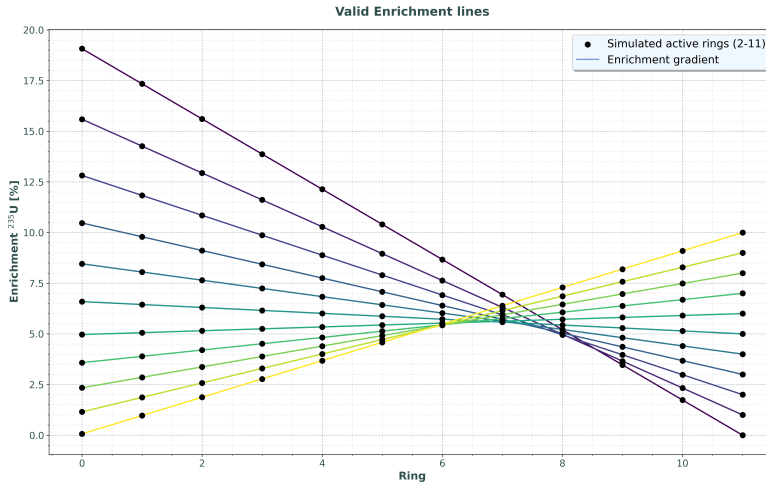
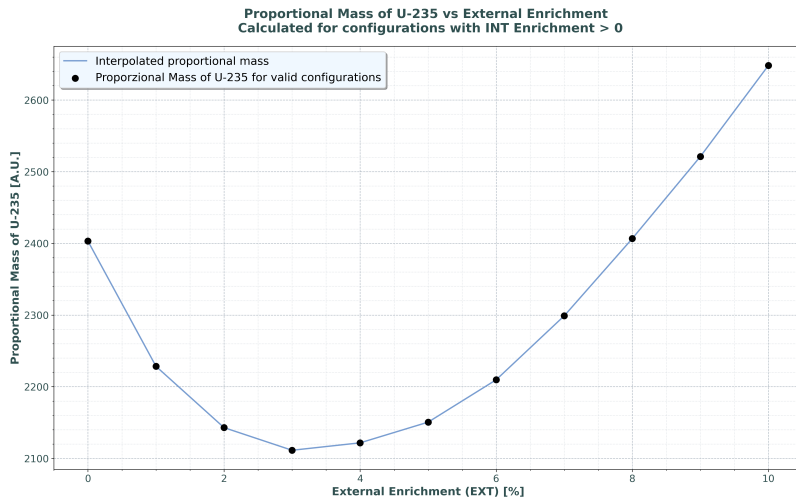
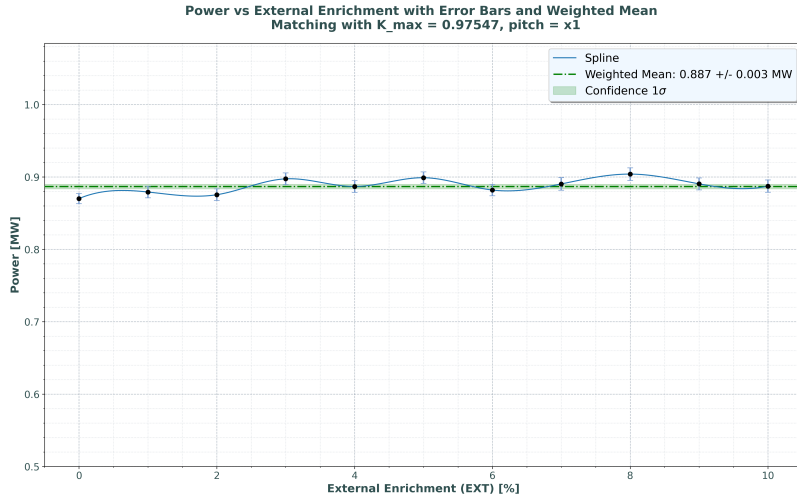


Figure: Radial extrapolation for the enrichment gradient.

# Extracted Power Variation

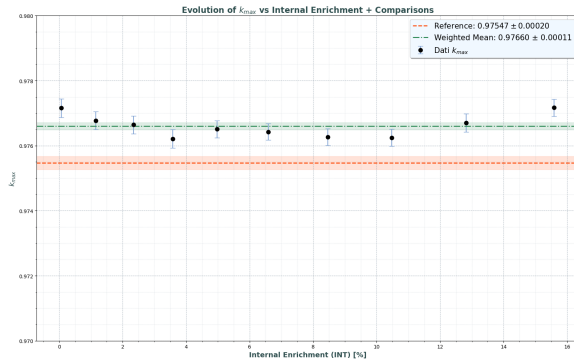


# Extracted Power Variation



# Model Consistency and Verification of $k_{max}$

- simulation across different linear enrichment configurations aimed at matching the reference  $k_{max}$ .
- Simulations performed on the three fundamental thermodynamic states: **CZP**, **HZP**, and **HFP**.
- The weighted average analysis confirms the substantial invariance of  $k_{max}$  values compared to the limiting case of homogeneous enrichment.
- This behavior formally validates the flexibility of the adopted geometric methodology.



# Performance Parameters: The Q-Factor

In an ADS system, the energy gain factor  $Q$  describes the amplification of the source power within the subcritical core:

$$Q = \frac{P_{thermal}}{P_{beam}} = \psi \frac{k_{eff}}{1 - k_{eff}}$$

Where  $\psi$  represents the relative importance factor of source neutrons.

For a consistent evaluation, a system  $Q$ -factor ( $Q_{sys}$ ) is defined with respect to the electrical power of the synchrotron:

$$Q_{sys} = \frac{P_{thermal}}{P_{el, sync} / \eta_{th}}$$

| Technology | Q-Value (Target/Record) | Regime                      |
|------------|-------------------------|-----------------------------|
| ITER       | $\sim 10$               | Fusion (Under construction) |
| SPARC      | $> 11$                  | Fusion (Under development)  |
| JET        | 0.67                    | Fusion (1997 Record)        |

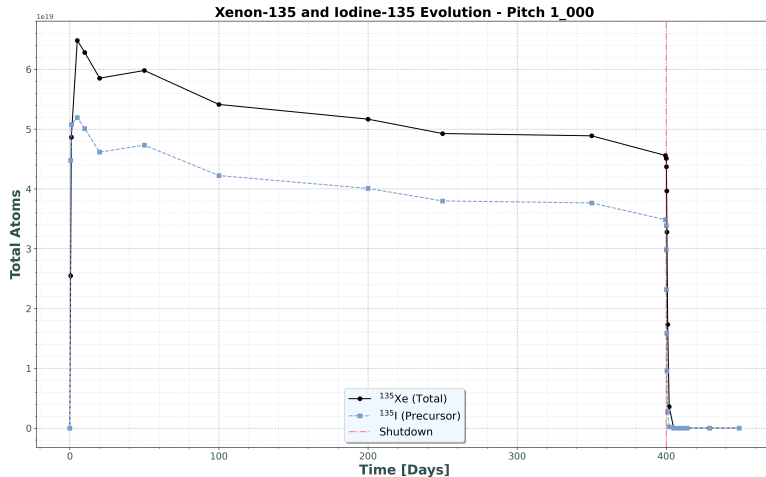
# Depletion Simulations (Burn-up)

## Transport-Isotopic Evolution Coupling

The simulation in *depletion* mode allows tracking the time evolution of the nuclear fuel isotopic composition under the effect of the neutron flux, by solving the coupled Monte Carlo transport and Bateman equations.

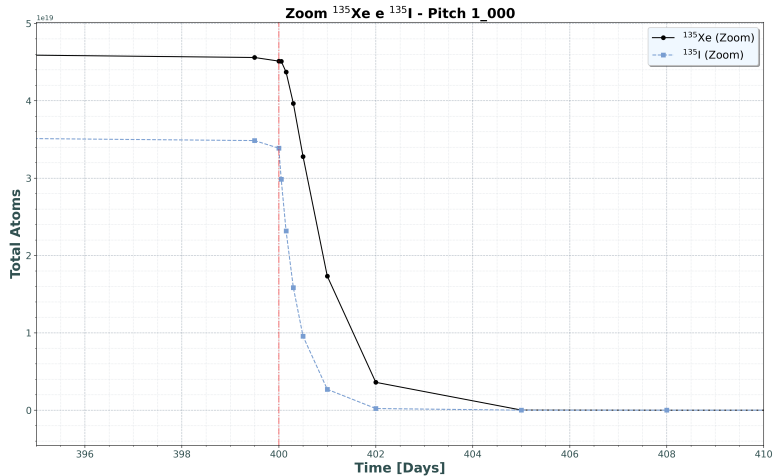
- **Transport Phase (OpenMC):** Calculation of the spatial and spectral neutron flux  $\Phi(\vec{r}, E)$  and microscopic reaction rates  $(\sigma_i \phi)$  at the beginning of the time-step.
- **Evolution Phase (Bateman):** Use of the extracted rates to numerically solve the system of linearized differential equations, updating the atomic densities  $N_i(t)$  for the next step.
- **Composition Feedback:** The accumulation of fission products (poisons) and the consumption of fissile material alter the macroscopic properties of the medium ( $\Sigma_a, \Sigma_f$ ), modifying the neutron response in subsequent steps.

# Time Evolution of $^{135}\text{I}$ and $^{135}\text{Xe}$



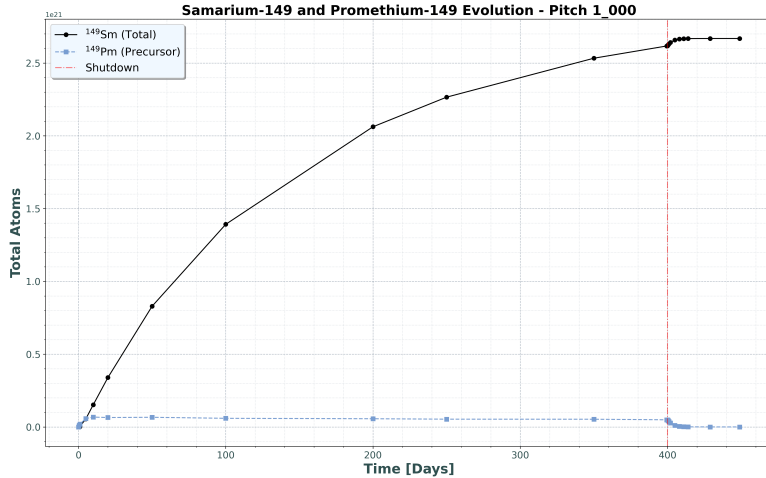
**Figure:** Coupled behavior of Iodine-135 and Xenon-135 atomic densities during nominal operation and during the post-shutdown phase.

# Time Evolution of $^{135}\text{I}$ and $^{135}\text{Xe}$



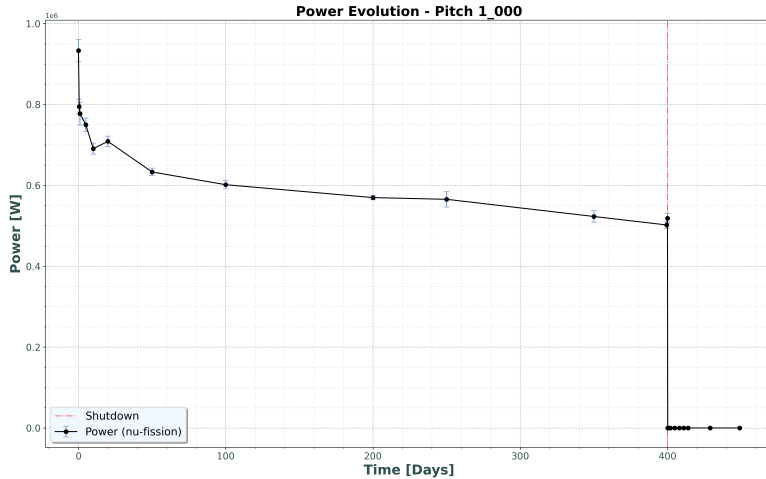
**Figure:** The graph highlights the onset of the post-shutdown poisoning peak induced by the decoupling of the half-lives of the two nuclides.

# Time Evolution of $^{149}\text{Sm}$



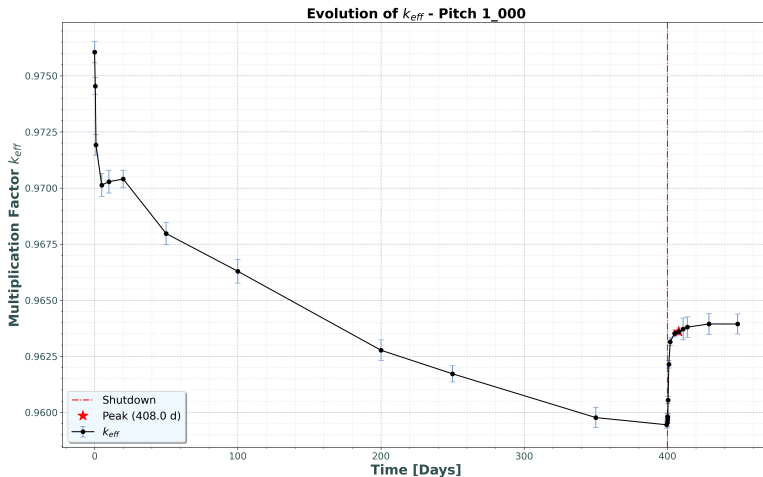
**Figure:** The graph highlights the asymptotic behaviour of samarium during core activity.

# Core Thermal Power Evolution



**Figure:** Time profile of the overall thermal power expressed in W over the fuel lifecycle.

# Evolution of the Effective Multiplication Factor ( $k_{eff}$ )



**Figure:** Trend of the effective multiplication factor  $k_{eff}$  as a function of Hot Full Power (HFP) operating days. The plot documents the legal safety margins and the overall stability of the system with respect to poisoning and fissile

# Mathematical Foundation: The Transport Equation

The steady-state neutron balance is governed by the transport equation:

## Operator Formulation

$$\mathcal{M}\Phi(\vec{r}, E, \hat{\Omega}) = \frac{1}{k_{eff}} \mathcal{F}\Phi(\vec{r}, E, \hat{\Omega})$$

The two operators describe the competing physical rates in the core:

- **$\mathcal{M}$  (Migration/Destruction Operator):** Quantifies net losses. It includes geometric leakage (*streaming*,  $\hat{\Omega} \cdot \nabla \Phi$ ), total interactions ( $\Sigma_t \Phi$ ), and kinematic repopulation due to scattering (*in-scattering*).
- **$\mathcal{F}$  (Production Operator):** Quantifies the steady-state source, describing the emission of new prompt neutrons induced by the integral fission rate ( $\chi \nu \Sigma_f \Phi$ ).

# Mathematical Foundation: The Krein-Rutman Theorem

Rewriting the equation as  $\mathcal{M}^{-1}\mathcal{F}\Phi = k_{eff}\Phi$ , the operator  $\mathcal{M}^{-1}\mathcal{F}$  maps one neutron generation to the next. Since atomic densities and cross-sections are strictly positive quantities, this operator is **rigorously positive**.

## Eigenvalue Dominance ( $k_{eff}$ )

By the **Krein-Rutman Theorem**, a positive integral operator guarantees the existence of a fundamental eigenvalue ( $k_0 \equiv k_{eff}$ ) that is real, positive, and **larger in magnitude** than all other eigenvalues in the spectrum ( $|k_0| > |k_n|$ ).

## Physical Meaning of the Eigenfunction ( $\Phi_0$ )

Only the dominant eigenvalue  $k_{eff}$  is associated with an eigenfunction  $\Phi_0$  that is **everywhere positive**. Higher-order harmonics inevitably exhibit nodes (negative fluxes, which are physically impossible in steady-state conditions).